Olympiad Circle Geometry, Power of a Point & Radical Axes

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Olympiad Circle Geometry, Power of a Point & Radical Axes Practice Exercises

1 In the circle shown, chords AB and CD intersect internally at point P. Given $PA = 4$, $PB = 9$, and $PC = 3$, find the exact segment length PD.

a. 10

b. 14

c. 12 ✓

d. 9

By the Intersecting Chords Theorem (Power of a Point inside a circle), $PA \cdot PB = PC \cdot PD$. Substituting gives $4 \cdot 9 = 3 \cdot PD \Rightarrow PD = \frac{36}{3} = 12$.

2 In the circle shown, chords AB and CD intersect internally at point P. Given $PA = 6$, $PB = 8$, and $PC = 4$, find the exact segment length PD.

a. 10

b. 12 ✓

c. 14

d. 9

By the Intersecting Chords Theorem (Power of a Point inside a circle), $PA \cdot PB = PC \cdot PD$. Substituting gives $6 \cdot 8 = 4 \cdot PD \Rightarrow PD = \frac{48}{4} = 12$.

3 In the circle shown, chords AB and CD intersect internally at point P. Given $PA = 5$, $PB = 12$, and $PC = 6$, find the exact segment length PD.

a. 10 ✓

b. 11

c. 12

d. 7

By the Intersecting Chords Theorem (Power of a Point inside a circle), $PA \cdot PB = PC \cdot PD$. Substituting gives $5 \cdot 12 = 6 \cdot PD \Rightarrow PD = \frac{60}{6} = 10$.

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4 In the circle shown, chords AB and CD intersect internally at point P. Given $PA = 6$, $PB = 10$, and $PC = 5$, find the exact segment length PD.

a. 11

b. 14

c. 9

d. 12 ✓

By the Intersecting Chords Theorem (Power of a Point inside a circle), $PA \cdot PB = PC \cdot PD$. Substituting gives $6 \cdot 10 = 5 \cdot PD \Rightarrow PD = \frac{60}{5} = 12$.

5 Cyclic quadrilateral $ABCD$ (shown schematically) has consecutive side lengths $AB = 3,\ BC = 4,\ CD = 3,\ DA = 4$, and diagonal $AC = 5$. By Ptolemy's Theorem, find the exact length of the other diagonal BD.

a. $\frac{7}{5}$

b. $\frac{24}{5}$

c. 5 ✓

d. 7

By Ptolemy's Theorem on cyclic quadrilateral $ABCD$, the product of the diagonals equals the sum of products of opposite sides: $AC \cdot BD = AB \cdot CD + BC \cdot DA = 3(3) + 4(4) = 25.$ Dividing by $AC = 5$ gives $BD = 5$.

6 Cyclic quadrilateral $ABCD$ (shown schematically) has consecutive side lengths $AB = 5,\ BC = 12,\ CD = 5,\ DA = 12$, and diagonal $AC = 13$. By Ptolemy's Theorem, find the exact length of the other diagonal BD.

a. $\frac{119}{13}$

b. 13 ✓

c. $\frac{120}{13}$

d. 15

By Ptolemy's Theorem on cyclic quadrilateral $ABCD$, the product of the diagonals equals the sum of products of opposite sides: $AC \cdot BD = AB \cdot CD + BC \cdot DA = 5(5) + 12(12) = 169.$ Dividing by $AC = 13$ gives $BD = 13$.

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7 Cyclic quadrilateral $ABCD$ (shown schematically) has consecutive side lengths $AB = 6,\ BC = 8,\ CD = 6,\ DA = 8$, and diagonal $AC = 10$. By Ptolemy's Theorem, find the exact length of the other diagonal BD.

a. 10 ✓

b. $\frac{14}{5}$

c. $\frac{48}{5}$

d. 12

By Ptolemy's Theorem on cyclic quadrilateral $ABCD$, the product of the diagonals equals the sum of products of opposite sides: $AC \cdot BD = AB \cdot CD + BC \cdot DA = 6(6) + 8(8) = 100$. Dividing by $AC = 10$ gives $BD = 10$.

8 Cyclic quadrilateral $ABCD$ (shown schematically) has consecutive side lengths $AB = 8,\ BC = 15,\ CD = 8,\ DA = 15$, and diagonal $AC = 17$. By Ptolemy's Theorem, find the exact length of the other diagonal BD.

a. $\frac{161}{17}$

b. $\frac{240}{17}$

c. 19

d. 17 ✓

By Ptolemy's Theorem on cyclic quadrilateral $ABCD$, the product of the diagonals equals the sum of products of opposite sides: $AC \cdot BD = AB \cdot CD + BC \cdot DA = 8(8) + 15(15) = 289$. Dividing by $AC = 17$ gives $BD = 17$.

9 In the Cartesian plane, circle ω_1 has equation $x^2 + y^2 = 33$ and circle ω_2 has equation $(x - 4)^2 + y^2 = 33$. Their radical axis (the locus of points having equal power with respect to ω_1 and ω_2) is the vertical line $x = x_0$. Find x_0.

a. 1

b. 4

c. 2 ✓

d. 3

Equating powers $(x^2 + y^2 - 33) = ((x - 4)^2 + y^2 - 33)$ cancels $x^2 + y^2$ and leaves $8x - 16 = 0 \Rightarrow 8x = 16 \Rightarrow x_0 = 2$.

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10 In the Cartesian plane, circle ω_1 has equation $x^2 + y^2 = 43$ and circle ω_2 has equation $(x-6)^2 + y^2 = 31$. Their radical axis (the locus of points having equal power with respect to ω_1 and ω_2) is the vertical line $x = x_0$. Find x_0.

a. 3

b. 4 ✓

c. 1

d. 5

Equating powers $(x^2 + y^2 - 43) = ((x-6)^2 + y^2 - 31)$ cancels $x^2 + y^2$ and leaves $12x - 36 = 12 \Rightarrow 12x = 48 \Rightarrow x_0 = 4$.

11 In the Cartesian plane, circle ω_1 has equation $x^2 + y^2 = 57$ and circle ω_2 has equation $(x-8)^2 + y^2 = 25$. Their radical axis (the locus of points having equal power with respect to ω_1 and ω_2) is the vertical line $x = x_0$. Find x_0.

a. 6 ✓

b. 4

c. 1

d. 7

Equating powers $(x^2 + y^2 - 57) = ((x-8)^2 + y^2 - 25)$ cancels $x^2 + y^2$ and leaves $16x - 64 = 32 \Rightarrow 16x = 96 \Rightarrow x_0 = 6$.

12 In the Cartesian plane, circle ω_1 has equation $x^2 + y^2 = 75$ and circle ω_2 has equation $(x-10)^2 + y^2 = 75$. Their radical axis (the locus of points having equal power with respect to ω_1 and ω_2) is the vertical line $x = x_0$. Find x_0.

a. 4

b. 10

c. 6

d. 5 ✓

Equating powers $(x^2 + y^2 - 75) = ((x-10)^2 + y^2 - 75)$ cancels $x^2 + y^2$ and leaves $20x - 100 = 0 \Rightarrow 20x = 100 \Rightarrow x_0 = 5$.

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